WILP, BITS Pilani

Session 13: Time Series Forecasting Methods

Course: Introduction to Statistical Methods (ISM) | Instructor: Dr. YVK Ravi Kumar

1 Moving Averages & Forecast Error Accuracy

Smooth out short-term fluctuations and evaluate forecast accuracy using MSE, MAD, MAPE.

Moving averages assume that future observations will resemble recent historical levels. Simple Moving Averages (SMA) weigh all k recent periods equally, whereas Weighted Moving Averages (WMA) assign larger weights to more recent observations. Forecast error is $e_t = y_t - F_t$.

Everyday Picture & Intuition

A simple moving average is like taking average temperature over the last 3 days to guess tomorrow's weather. Accuracy metrics like MSE (Mean Squared Error) and MAD (Mean Absolute Deviation) measure how far off your guesses are on average.

1.1 Mathematical Formulas

$$\mathrm{SMA}_t = \frac{y_{t-1} + y_{t-2} + \cdots + y_{t-k}}{k}, \quad \mathrm{WMA}_t = \frac{w_1 y_{t-1} + w_2 y_{t-2} + \cdots + w_k y_{t-k}}{\sum w_i} \tag{1}$$

$$\mathrm{MSE} = \frac{1}{n}\sum (y_t - F_t)^2, \quad \mathrm{MAD} = \frac{1}{n}\sum |y_t - F_t|, \quad \mathrm{MAPE} = \frac{100\%}{n}\sum \left|\frac{y_t - F_t}{y_t}\right| \tag{2}$$

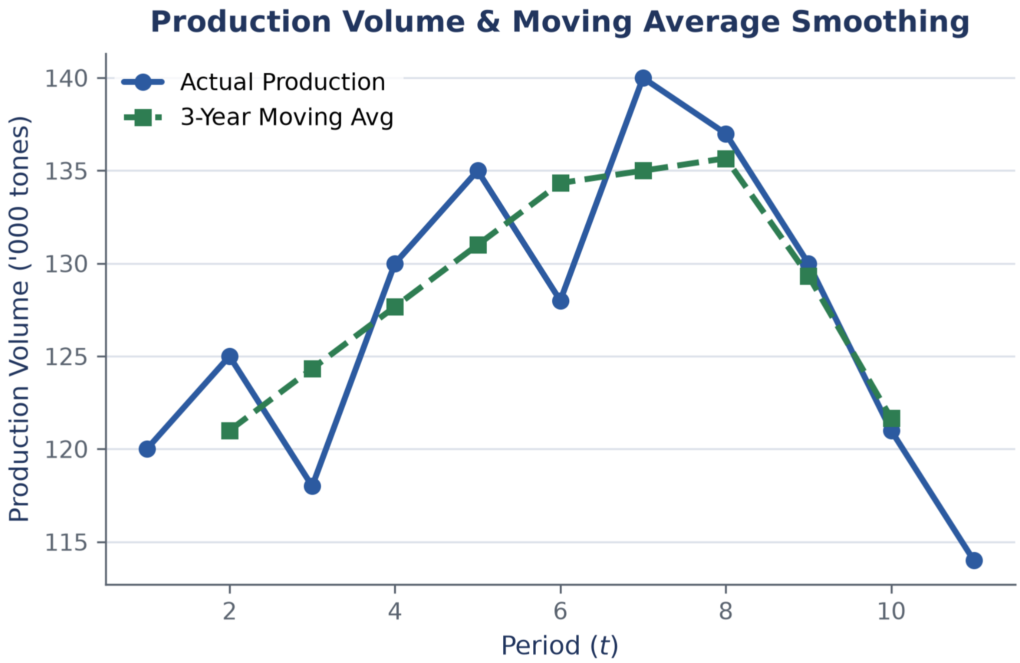

Figure 1: Comparison of raw production data against a 3-period moving average curve.

Fully Worked Example

Demand observations for recent months: Month 49 = 130, Month 50 = 128, Month 51 = 137.

Step 1: Compute 3-month Simple Moving Average forecast for Month 52:

$$F_{52} = \frac{137 + 128 + 130}{3} = \frac{395}{3} = 131.67 \tag{3}$$

Step 2: Compute Weighted Moving Average with weights $w = [3, 2, 1]$ (highest weight on most recent period):

$$F_{52} = \frac{3(137) + 2(128) + 1(130)}{3 + 2 + 1} = \frac{411 + 256 + 130}{6} = \frac{797}{6} = 132.83 \tag{4}$$

2 Simple Exponential Smoothing

Assign exponentially decaying weights to past historical observations.

Simple Exponential Smoothing (SES) updates forecasts using a smoothing constant $\alpha \in [0, 1]$:

$$F_{t+1} = \alpha y_t + (1 - \alpha)F_t = F_t + \alpha(y_t - F_t) \tag{5}$$

Everyday Picture & Intuition

Exponential smoothing is like updating your guess for commute time. Your new guess equals your old guess plus a fraction α of your error from today's commute.

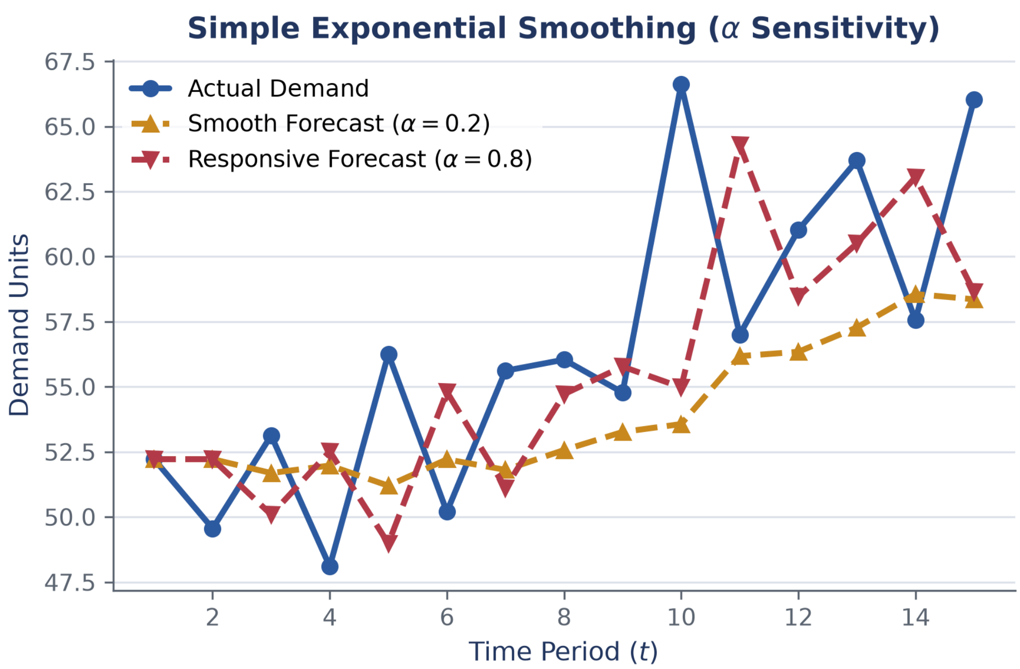

Figure 2: Sensitivity of exponential smoothing forecasts under different smoothing constants $\alpha = 0.2$ vs $\alpha = 0.8$.

Fully Worked Example

Suppose forecast for the first week of March was $F_1 = 500$ units, but actual demand was $y_1 = 450$ units. Let $\alpha = 0.3$.

Step 1: Calculate forecast for week 2:

$$F_2 = 0.3(450) + (1 - 0.3)(500) = 135 + 350 = 485 \tag{6}$$

Step 2: Suppose actual demand in week 2 is $y_2 = 505$ units. Calculate forecast for week 3:

$$F_3 = 0.3(505) + 0.7(485) = 151.5 + 339.5 = 491.0 \tag{7}$$

3 Holt's Double Exponential Smoothing

Track both baseline level and linear trend simultaneously in trending series.

When data exhibits a persistent linear trend, simple exponential smoothing lags behind. Holt's Double Exponential Smoothing introduces a second smoothing equation to update the trend component T_t alongside level L_t.

Everyday Picture & Intuition

If sales grow by 10 units every month, simple smoothing will always underestimate next month's sales. Holt's method tracks both current height and growth speed.

3.1 Holt's Equations

$$L_t = \alpha y_t + (1 - \alpha)(L_{t-1} + T_{t-1}) \tag{8}$$
$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \tag{9}$$
$$\hat{y}_{t+h} = L_t + hT_t \tag{10}$$

Watch Out! Pitfalls

Simple moving averages and single exponential smoothing are only suitable for stationary series. For trending data, use Holt's method or ARIMA.

Key Takeaways

- Simple Moving Averages average recent k values equally; WMAs emphasize recent data.
- Error metrics MSE, MAD, and MAPE measure forecast accuracy.
- Exponential Smoothing updates forecasts by adding a fraction α of past forecast error.
- Holt's Double Exponential Smoothing incorporates a trend factor to eliminate lag on trending data.

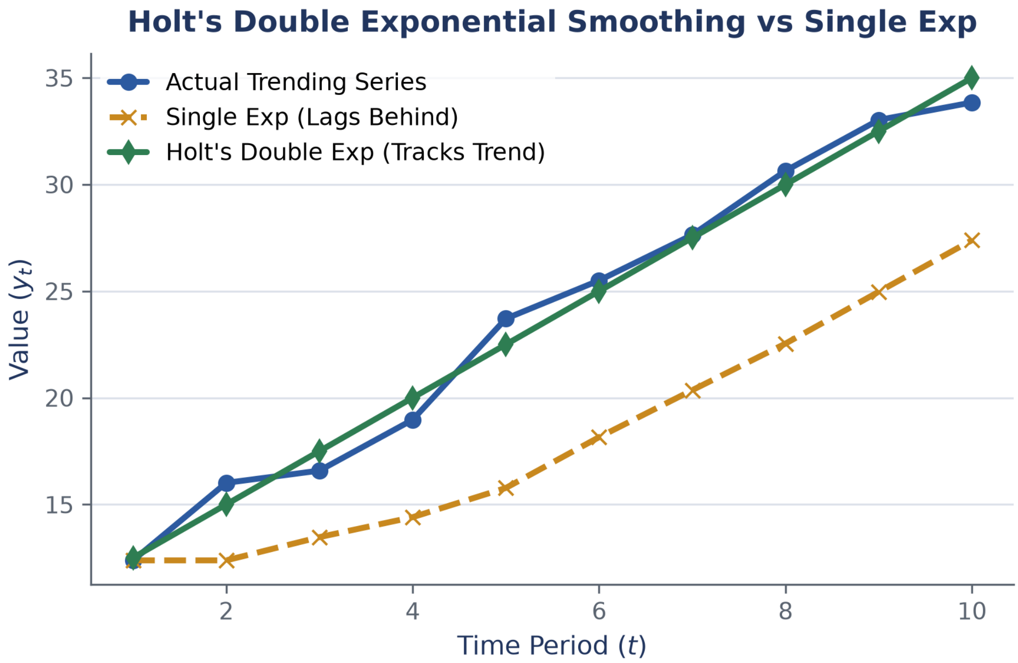

Figure 3: Holt's double exponential smoothing tracking trending time series without lagging.